Innocent Kisoka

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Machine Learning / AI — generative models, evaluation, SciML, HPC

PROFILE

Machine learning / AI (M.Sc. Artificial Intelligence, USI, Switzerland — graduated 26 September 2026; B.Sc. Computer Engineering, Kyungdong University, South Korea). Builds and evaluates learning systems carefully: LoRA/PEFT LM adaptation, multi-seed metrics, structure-visible SciML, and CUDA/HPC experiment software. Evaluation-first — when scores look fine but behaviour is off, redesign checks before changing the model. Comfortable moving from small reproducible experiments to multi-GPU runs. English C1 (thesis in English).

TECHNICAL SKILLS

ML / generative AI	PyTorch, LoRA/PEFT, preference-style training, multi-seed evaluation, Transformers
SciML / structure	FEM+PINN scientific ML, GNNs (PyG), topology-aware overfitting checks
Systems / HPC	CUDA C++, multi-GPU CSCS pipelines, MPI+CUDA, reproducible experiments
Core	Python, Git, Linux, scientific writing (EN). Generative AI with LLMs (skills)
Spoken	English C1 • Italian (working) • Korean B2 • Swahili (native)

EDUCATION

M.Sc. Artificial Intelligence Sep 2024 – 26 Sep 2026

Università della Svizzera italiana (USI), Lugano, Switzerland — officially graduated 26 September 2026

- Coursework: Deep Learning, Scientific Machine Learning, HPC, Advanced Topics in ML, Computer Vision, NLP.
- Thesis: multilingual LM preference optimisation — LoRA/PEFT, multi-seed metrics, multi-GPU at CSCS.

CSCS–USI Summer School: Effective HPC & Data Analytics Jul 2026

Swiss National Supercomputing Centre (CSCS), Switzerland — CUDA, GPU architectures, MPI+CUDA.

B.Sc. Computer Engineering Sep 2020 – Aug 2024

Kyungdong University, South Korea.

PROJECTS

LLM adaptation & evaluation (Master's thesis) 2024 – 2026

- Closed-loop evaluation caught behaviour failures when automatic scores looked fine.
- Multi-seed LoRA/PEFT preference-optimisation experiments on multi-GPU pipelines at CSCS.

Cardiac diffusion PDE — FEM + PINN (SciML) 2025 – 2026

- Couples FEM structure with PINN-style learning so the model stays physics-visible rather than a black box.

CUDA / HPC numerical software 2025 – 2026

- Kernels, stencils, async pipelines, PDE mini-apps — experiment software for serious numerical runs.

GNN replication — topological overfitting 2025 – 2026

- Reproduced structure-aware checks in PyG and documented topology-driven overfitting.

EXPERIENCE

Teaching Assistant — Web Development Feb – Jun 2023

Kyungdong University, South Korea

- Supported labs and mentoring for undergraduate web-development coursework.

AWARDS

2024 — Scholarship, Foundation for the Lugano Faculties of USI • 2022 — Programming Excellence Award, Kyungdong University • 2020 — KDU Global–IUCEA 100% Scholarship